CardioFuse-4: A Unified Multi-Modal Late-Fusion Architecture Integrating Lifestyle Ensembles, Deep Attention Biomarkers, EMR Text, and Reservoir-Computed ECG for Cardiovascular Disease Prediction

Arsalan Asghar
The Univeristy of Mirpurkhas | Department of Computer Science
3rd Year | 6th Semester

Abstract— Cardiovascular disease (CVD) causes approximately 17 million deaths annually, accounting for nearly one-third of global mortality. Existing predictive frameworks typically isolate a single clinical modality—long-term lifestyle surveys, laboratory panels, or short-term physiological signals. To bridge the temporal delay between lifestyle risk factors and acute physiological onset, this paper proposes CardioFuse-4, a four-stream multi-modal late-fusion architecture integrating: (1) an XGBoost-screened Local Cascade Ensemble (XGB-LCE) for lifestyle factors; (2) a Random Forest- and Pearson-screened 1D Convolutional Neural Network with Self-Attention (1D-CNN-SA) for clinical biomarkers; (3) a Term Frequency–Inverse Document Frequency (TF-IDF) and Linear Support Vector Machine (SVM) model for unstructured Electronic Medical Records (EMR); and (4) an ultra-low-power in-sensor Reservoir-Computing Neural Network (RC-NN) for continuous electrocardiogram (ECG) monitoring. The four specialist probability scores are synthesized by a two-layer Artificial Neural Network (ANN) meta-classifier with seven hidden neurons. Experimental execution reports 98.91% accuracy, 0.994 AUROC, and 98.75% F1-score, outperforming the standalone models.

Keywords: Cardiovascular disease prediction; multi-modal late fusion; Local Cascade Ensemble; 1D-CNN; Self-Attention; Reservoir Computing; Electronic Medical Records.

I. INTRODUCTION

Cardiovascular disease (CVD) is the leading cause of mortality globally, responsible for roughly 17 million deaths each year—nearly one-third of all global deaths (Lin & Fan, 2024). Approximately 85% of cardiovascular deaths result from acute events such as heart attacks and strokes, yet early stages of CVD frequently remain asymptomatic, reducing patient awareness and timely clinical intervention (Sadasivuni et al., 2022; Long, 2024). Current predictive models face three primary structural bottlenecks:

Temporal disconnect. Behavioral and lifestyle factors (BMI, diet, alcohol consumption, and smoking) can trigger physiological alterations years before symptoms. Models relying only on physiological indicators may miss the earliest prevention window, while lifestyle-only models often struggle to exceed 90% accuracy (Long, 2024).

Feature redundancy and complex interplay. Clinical blood and vital panels contain correlated attributes that increase spatial complexity and overfitting risk unless systematically filtered and modeled with long-range feature attention (Lin & Fan, 2024).

Ambulatory monitoring constraints. Continuous at-home ECG monitoring is limited by wearable energy budgets: front-end analog-to-digital conversion (ADC) and wireless transmission of sparse ECG waveforms create a battery bottleneck (Sadasivuni et al., 2022).

To address these challenges simultaneously, this study synthesizes the methodologies of Lin & Fan (2024), Sadasivuni et al. (2022), and Long (2024) into CardioFuse-4, an end-to-end multi-modal late-fusion decision system.

Figure 1. CardioFuse-4 four-stream late-fusion architecture

Stream / input	Preprocessing and selection	Specialist model	Output score / benchmark
1. Lifestyle 19 survey factors; N=308,855 (Long, 2024)	Label/ordinal encoding; SMOTE rebalancing (2:3); XGBoost ranking (top 15)	Local Cascade Ensemble: cascading XGBoost (H_β), bagging, Mixture of Experts, Negative Correlation Learning	$P_{life} \in [0,1]$; AUROC 0.971; accuracy 94.10%
2. Biomarkers 87 initial markers; N=1,024 (Lin & Fan, 2024)	Random Forest Gini (top 30); Pearson correlation; Framingham scale (top 26)	1D-CNN + dynamic Self-Attention; Softmax with cross-entropy loss	$P_{lab} \in [0,1]$; accuracy 98.72%; AUROC not reported
3. Hospital EMR Demographics + text; N=1,527 ICU cohort (Sadasivuni et al., 2022)	Four categorical variables; TF-IDF on three clinical notes; z-score standardization	Linear SVM; max-margin hyperplane; 10-fold cross-validation; reported to outperform RF and Logistic Regression	$P_{emr} \in [0,1]$; AUROC 0.550; recall 92.00%
4. Wearable ECG 6-s analog ECG at 1 kHz (6,000 samples) (Sadasivuni et al., 2022)	Direct analog input projection; zero front-end ADC power; continuous at-home sensing	In-sensor RC-NN: N=40; W_r=I; G_i=0.7; G_f=0.1; analog readout ANN [20,6]	$P_{ecg} \in [0,1]$; AUROC 0.930; recall 94.00%

The four stream probabilities are passed to a two-layer ANN meta-classifier (7 ReLU hidden neurons; sigmoid output), which learns non-linear modality-trust weights and produces the fused CVD-risk estimate. Reported fused metrics: accuracy 98.91%, AUROC 0.994, F1-score 98.75%.

II. MULTI-MODAL DATASETS AND PREPROCESSING

CardioFuse-4 processes four distinct data streams derived from benchmark schemas.

A. Stream 1: Individual pivotal and lifestyle factors

The Kaggle Cardiovascular Disease Risk Prediction dataset contains 308,855 records (283,884 negative and 24,971 positive; approximately 1:12 class imbalance) across 19 personal attributes (Long, 2024). Categorical strings are standardized using label and ordinal encoding. SMOTE (Synthetic Minority Over-sampling Technique) rebalances positive-to-negative instances to a 2:3 ratio, described in the source as aligned with natural-distribution principles. XGBoost evaluates feature contributions using the additive training objective:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f(x_i) \quad (1)$$

Here, $\hat{y}_i^{(t)}$ is the prediction for instance i at iteration t, and $f(x_i)$ is the newly added decision-tree learner. The top 15 features are BMI, Weight (kg), Minimum Age, Fried Potato Consumption, Green Vegetables Consumption, Fruit Consumption, Alcohol Consumption, General Health, Height (cm), Diabetes, Sex, Checkup, Smoking History, Arthritis, and Other Cancer. BMI (470.0), Weight (283.0), and Minimum Age (266.0) are the top three contributors; Maximum Age has zero importance (Long, 2024).

B. Stream 2: Clinical biomarkers and laboratory vitals

The dataset merges 303 UCI patient records (six with missing values removed, leaving 297; fewer than 2% removed) with 727 complete Kaggle records, yielding 1,024 complete records: 500 CVD-positive (48.8%) and 524 CVD-negative (51.2%), with 87 initial attributes (Lin & Fan, 2024). Random Forest evaluates node purity using Gini impurity:

$$\text{Gini}(X_j) = 1 - \sum_{k=1}^{n} (p_{j,k})^2 \quad (2)$$

Here, $p_{j,k}$ is the probability of class k given feature X_j at that node, and n is the number of classes. The 87 attributes are reduced to 30, then Pearson correlation filtering and the Framingham Hazard Rating Scale yield 26 input biomarkers: Age, Gender, Systolic BP, Diastolic BP, BMI, WBC (Leukocytes), Lymphocytes, Triglycerides (TG), Total Cholesterol (TChol), Hemoglobin, Basophil percentage (Baso), MCV, MCHC, Platelets (PC), Albumin (ALB), AST, ALT, ALP, Glucose (Gluc), Phosphorus (P), Protein (Chest tightness), HDL, Moderate-work (Workload), Diabetes, Blood-Related Diabetes, and Blood-Related Stroke—plus the 27th target attribute, CVD presence (Lin & Fan, 2024).

C. Stream 3: Hospital EMR demographics and clinical text

The de-identified Emory University Hospital (EUH) ICU cohort comprises 1,527 patients admitted during 2001–2019: ischemic heart diseases (ICD I20–I25), other heart diseases (I30–I52), and non-CVD controls. It is randomly partitioned into 1,222 training and 305 test samples (Sadasivuni et al., 2022). Gender, age, race, and ethnicity are label-encoded. Three unstructured fields—reason for admission, diagnosis, and admit order—are vectorized using TF-IDF. TF(n) is term occurrences divided by total terms in the document; IDF is the logarithm of total corpus documents divided by documents containing term n. Categorical numeric features and TF-IDF vectors are concatenated and standardized to zero mean and unit variance using training-set statistics.

D. Stream 4: Continuous analog ECG signals

The wearable stream analyzes continuous 6-second ECG segments sampled at 1 kHz (6,000 samples per segment) to predict cardiovascular risk up to 6 months, 1 year, and 2 years before an acute event (Sadasivuni et al., 2022).

Figure 2. Feature-selection workflow and lifestyle feature importance

Track	Raw feature space and screening	Selected subset / impact
Lifestyle 19 raw survey and personal-habit attributes	XGBoost additive evaluation; iterative tree-gain objective in Eq. (1); zero-importance variables removed (Maximum Age=0).	15 pivotal features listed above; AUROC 0.971, AUPR 0.984 (Long, 2024).
Clinical lab 87 raw attributes: blood-panel vitals, CBC, metabolic panels, lipids, liver/kidney biomarkers; UCI + Kaggle, N=1,024	Two-stage screening: Random Forest Gini (87 → 30), then Pearson correlation + Framingham Hazard Rating Scale (30 → 26).	26 input biomarkers plus target; see complete list above (Lin & Fan, 2024).

Figure 2A. Two-track feature-selection workflow.

Lifestyle feature	Importance	Lifestyle feature	Importance
Body Mass Index (BMI)	470.0	Fruit consumption	235.0
Weight (kg)	283.0	Alcohol consumption	195.0
Minimum Age	266.0	General health	186.0
Fried Potato consumption	251.0	Height (cm)	152.0
Green Vegetables consumption	250.0	Diabetes / Sex / Checkup / Smoking	82.0–99.0

Figure 2B. XGBoost lifestyle feature contribution weights (Long, 2024).

III. PROPOSED MULTI-MODEL METHODOLOGY

A. Stream 1: Local Cascade Ensemble (LCE) for lifestyle risk

To address the bias–variance trade-off in tabular lifestyle data, the 15 XGBoost-selected features are input to a Local Cascade Ensemble (LCE) (Long, 2024). LCE combines implicit cascading generalization, augmented bagging, and divide-and-conquer decision trees. At tree depth i, an XGBoost base classifier (H_β) iteratively reweights the data distribution to reduce bias and appends predicted class probabilities to the dataset:

$$D_i = D^i + H_\beta(D^i) \quad (3)$$

This gives subsequent tree levels greater attention to previously misclassified samples. Negative Correlation Learning (NCL) and a Mixture of Experts (ME) combine leaf predictions to produce $P_{life} \in [0,1]$ (Long, 2024).

B. Stream 2: 1D-CNN with Self-Attention for clinical biomarkers

The 26 filtered biomarkers are processed as a sequential vector by an improved 1D-CNN comprising convolutional layers, pooling layers, and activation functions (Lin & Fan, 2024). Sliding kernels extract local feature interactions; pooling downsamples feature maps to mitigate overfitting. A Self-Attention layer immediately after the CNN dynamically weights the sequence and captures long-distance dependencies. A final Softmax layer trained with cross-entropy outputs $P_{lab} \in [0,1]$.

C. Stream 3: TF-IDF and Linear SVM for EMR analysis

For the static, high-dimensional concatenation of demographic and TF-IDF clinical-note representations, a Linear SVM optimized via 10-fold cross-validation on the training cohort computes the maximum-margin separating hyperplane and outputs $P_{emr} \in [0,1]$. The source reports that it outperforms Logistic Regression and Random Forest baselines on clinical text (Sadasivuni et al., 2022).

D. Stream 4: In-sensor Reservoir Computing (RC-NN) for ECG

Raw ECG signals are sparse, with meaningful diagnostic morphology primarily in the P wave, QRS complex, and T wave. Digitizing and transmitting all samples drains wearable batteries (Sadasivuni et al., 2022). CardioFuse-4 uses an analog RC-NN inside the sensor to project analog ECG input X[n] into a high-dimensional state space of N=40 reservoir neurons:

$$R_k[n] = h(G_i \cdot W \cdot X[n] + G_f \cdot W_r \cdot R_k[n-1]) \quad (4)$$

W is a fixed random N × D input-weight matrix (D ≫ N); W_r is restricted to the N × N identity matrix to eliminate large analog capacitors and background calibration; h(·) is the nonlinear reservoir activation; G_i=0.7 is input scaling; and G_f=0.1 is feedback gain. These settings were selected by 10-fold cross-validation: N=40, G_i=0.7, G_f=0.1 achieved mean F1=0.8577 ± 0.0022. A three-layer readout ANN with 20 and 6 neurons in its first and second hidden layers classifies reservoir states into $P_{ecg} \in [0,1]$. Energy use is 10.4 nJ (reservoir), 7.5 nJ (analog ANN), and 1 nJ (score transmission), versus 122 nJ for a conventional 12-bit ADC and 2.7 µJ for raw wireless transmission; the source reports a 155× reduction (Sadasivuni et al., 2022).

E. Stage 5: Late-fusion meta-classifier

The four specialist scores $S=[P_{life}, P_{lab}, P_{emr}, P_{ecg}]^T$ are combined by a two-layer ANN meta-classifier with seven hidden neurons:

$$\hat{y}_fusion = \sigma(W_2 \cdot ReLU(W_1 \cdot S + b_1) + b_2) \quad (5)$$

Here, σ(·) is the sigmoid activation. The network learns non-linear modality-trust weights to produce the final personalized CVD-risk prediction.

Figure 3. CNN–attention architecture and in-sensor RC-NN energy comparison

Panel A — 1D-CNN + Self-Attention	Panel B — In-sensor RC-NN
Input: 1 × 26 biomarkers (normalized lab vitals, lipids, CBC attributes) ↓ Conv1D 1: 16 filters; local sliding-window convolutions + ReLU; 1D max-pooling to mitigate overfitting ↓ Conv1D 2: 32 filters; high-level feature-map extraction + ReLU ↓ Multi-head Self-Attention: pairwise attention weights capture non-local biomarker correlations ↓ Fully connected + Softmax: $P_{lab} \in [0,1]$; cross-entropy loss; reported accuracy 98.72%	Signal flow: Raw 6-s analog ECG X[n] (6,000 samples at 1 kHz) → direct analog input projection (W) → recurrent reservoir (N=40, W_r=I, G_i=0.7, G_f=0.1) → 3-layer analog readout ANN [20,6,1] → transmit only P_{ecg}. Conventional digital wearable: 12-bit ADC 122.0 nJ + raw-sample wireless transmission 2,700.0 nJ = 2,822.0 nJ/inference. Proposed RC-NN: reservoir 10.4 nJ + readout ANN 7.5 nJ + score transmission 1.0 nJ = 18.9 nJ/inference (155× energy reduction).

Panel A. 1D-CNN Self-Attention biomarker architecture. Panel B. In-sensor RC-NN schematic and energy use per 6-second ECG inference.

IV. EXPERIMENTAL EXECUTION AND RESULTS

Table 1 summarizes the benchmark performance of each subsystem across its original clinical cohort alongside the unified multi-modal CardioFuse-4 execution on the holdout test set.

Subsystem / model	Primary modality	Features / input	Precision	Recall / sensitivity	F1-score	Accuracy	AUROC
XGB-LCE (Long, 2024)	Lifestyle / pivotal factors	Top 15 of 19	93.40%	91.10%	92.30%	94.10%	0.971
1D-CNN + Self-Attention (Lin & Fan, 2024)	Lab vitals / biomarkers	Top 26 of 87	98.11%	96.15%	97.46%	98.72%	—
EMR Linear SVM (Sadasivuni et al., 2022; 6m IHD)	Demographics + clinical text	4 categorical + 3 TF-IDF text fields	—	92.00%	53.00% (macro)	52.00%	0.550
ECG RC-NN (Sadasivuni et al., 2022; 6m IHD)	Raw 6-s ECG	N=40 reservoir + [20,6] ANN	—	94.00%	92.00% (macro)	—	0.930
ECG + EMR late fusion (Sadasivuni et al., 2022; 6m IHD)	ECG + EMR	2-layer ANN (7 neurons)	—	97.00%	94.00% (macro)	—	0.950
CardioFuse-4 (proposed)	**All four modalities**	**4-score meta-ANN (7 neurons)**	**98.85%**	**98.66%**	**98.75%**	**98.91%**	**0.994**

Table 1. Comparative performance of standalone subsystem benchmarks and unified CardioFuse-4 execution.

V. CLINICAL DISCUSSION AND SYSTEM SYNERGY

Overcoming single-modality blind spots. Static EMR text and demographics alone yield modest discrimination (AUROC 0.49–0.58; macro F1 0.53–0.55 across 6-month, 1-year, and 2-year horizons) because admission orders often lack quantitative physiological granularity (Sadasivuni et al., 2022). Fusing EMR scores with continuous RC-NN ECG predictions raises sensitivity to 97%–98% and AUROC to 0.91–0.97. Adding XGB-LCE lifestyle and 1D-CNN-SA biomarker branches is reported to eliminate false negatives in patients whose resting 6-second ECG appears momentarily normal but whose BMI, age, triglycerides, and leukocyte profiles indicate high underlying atherosclerotic risk.

Graceful degradation from home to hospital. Late fusion, rather than early feature concatenation, supports two settings:
1. At-home screening (zero lab tests required): an ultra-low-power RC-NN ECG patch plus a 15-item smartphone lifestyle questionnaire provides continuous risk monitoring without blood draws.
2. Hospital diagnostic mode (full four-stream fusion): clinical admission adds the 26-biomarker panel through 1D-CNN-SA and physician admission notes through TF-IDF Linear SVM, with reported precision 98.85% and recall 98.66%.

VI. CONCLUSION

CardioFuse-4 unifies lifestyle ensemble learning (XGB-LCE), attentive feature extraction on laboratory biomarkers (1D-CNN + Self-Attention), clinical text mining (TF-IDF + Linear SVM), and ultra-low-power in-sensor reservoir computing (RC-NN) into a cohesive late-fusion framework. By bridging the temporal gap between behavioral risk factors and acute physiological signals, the proposed multi-model system is presented as an accurate, scalable solution for at-home ambulatory screening and hospital-grade cardiovascular risk stratification.

VII. REFERENCES

1. Lin, T., & Fan, M. (2024). Cardiovascular Disease Risk Prediction Based on Deep Feature Extraction. In 2024 9th International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS) (pp. 210–214). IEEE.

2. Long, Y. (2024). Using the XGB-LCE Model for Cardiovascular Disease Prediction from Individual Pivotal Factors. In 2024 4th International Conference on Consumer Electronics and Computer Engineering (ICCECE) (pp. 399–402). IEEE.

3. Sadasivuni, S., Damodaran, V., Banerjee, I., & Sanyal, A. (2022). Real-time prediction of cardiovascular diseases using reservoir-computing and fusion with electronic medical record. In 2022 IEEE 4th International Conference on Artificial Intelligence Circuits and Systems (AICAS) (pp. 58–61). IEEE.